HANDOUT OPTICAL INSTRUMENTS

Optical Instruments

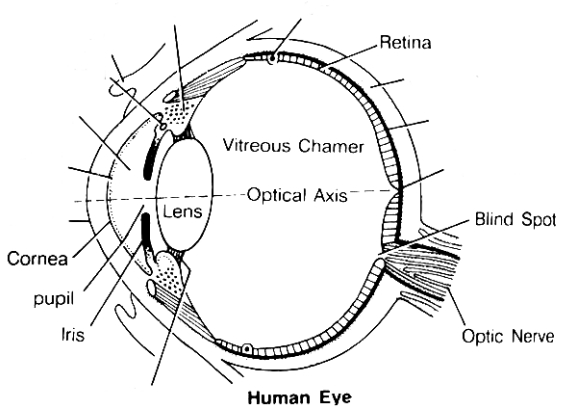
1 Human Eye

1.1 Structure of Eye

Light enters the eye through a curved front surface, the cornea. It passes through the pupil which is the central hole in the iris. The size of the pupil can change under control of muscles. The light is further focussed by the eye-lens on the retina. The retina is a film of nerve fibres covering the curved back surface of the eye. The retina contains rods and cones which sense light intensity and colour, respectively, and transmit electrical signals via the optic nerve to the brain which finally processes this information. The shape (curvature) and therefore the focal length of the lens can be modified somewhat by the ciliary muscles. For example, when the muscle is relaxed, the focal length is about 2.5 cm and (for a normal eye) objects at infinity are in sharp focus on the retinas.

When the object is brought closer to the eye, in order to maintain the same image-lens distance (2.5 cm), the focal length of the eye-lens becomes shorter by the action of the ciliary muscles. This property of the eye in called **accommodation**.

If the object is too close to the eye, the lens cannot curve enough to focus the image on to the retina, and the image is blurred.



Human Eye

The closest distance for which the lens can focus light on the retina is called the **least distance of distinct vision or the near point.** The standard value (for normal vision) taken here is 25 cm (the near point is given the symbol D.)

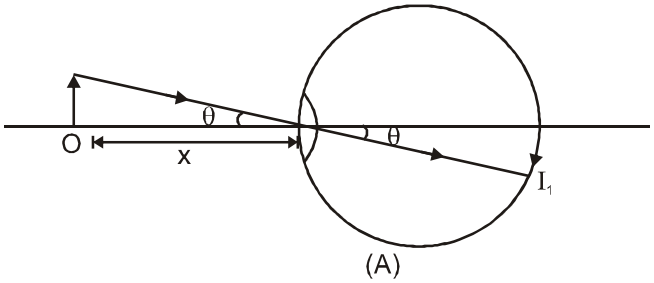
When the image is situated at infinity the ciliary muscles are least strained to focus the final image on the retina, this situation is known as **normal adjustment**.

1.2 Regarding Eye:

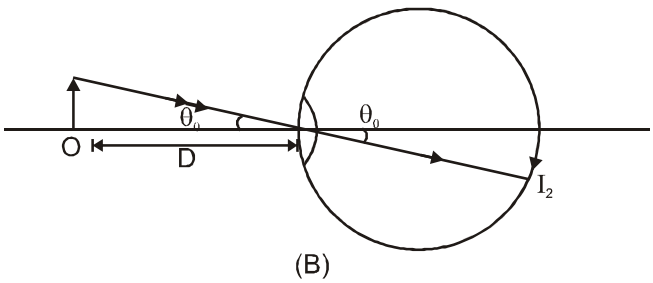
1. In eye convex eye-lens forms real inverted and diminished image at the retina by changing its convexity (the distance between eye lens and retina is fixed)

2. The human eye is most sensitive to yellow green light having wavelength 5550 Å and least to violet (4000Å) and red (7000 Å)

3. The size of an object as perceived by eye depends on its visual-angle when object is distant its visual angle θ and hence image I_1 at retina is small (it will appear small) and as it is brought near to the eye its visual angle θ_0 and hence size of image I_2 will increase.



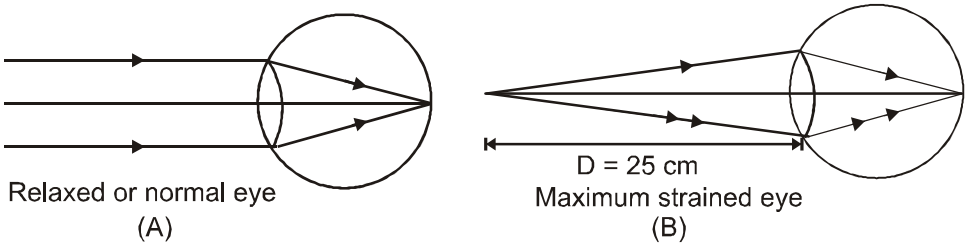
(A)



(B)

4. The far and near point for normal eye are usually taken to be infinity and 25 cm respectively i.e.., normal eye can see very distant object clearly but near objects only if they are at distance greater than 25 cm from the eye. The ability of eye to see objects from infinite distance to 25 cm from it is called **Power of accommodation.**

5. If object is at infinity i.e. parallel beam of light enters the eye is least strained and said to be relaxed or unstrained. However, if the object is at least distance of distinct vision (L.D.D.V) i.e., D (=25 cm) eye is under maximum strain and visual angle is maximum.



Relaxed or normal eye (A)

Maximum strained eye (B)

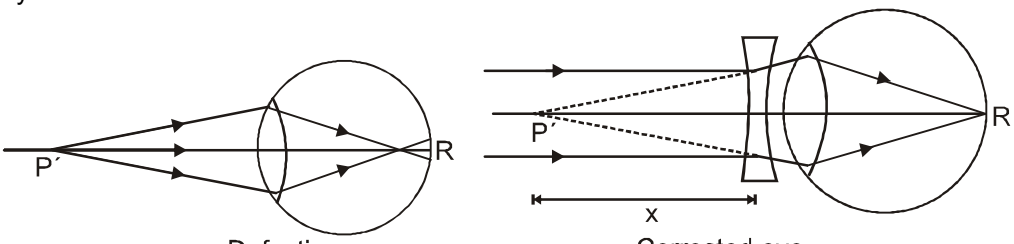
6. The limit of resolution of eye is one minute i.e. two object will not be visible distinctly to the eye if the angle subtended by them on the eye is lesser than one minute.

7. The persistence of vision is (1/10) sec i.e., If time interval between two consecutive light pulses is lesser than 0.1 sec eye cannot distinguish them separately. This fact is taken into account in motion pictures.

1.3 Defects of vision

1. Myopia [or short-sightendness or near - sightendness]

In it distant objects are not clearly visible. The far point for a myopic eye is much nearer than infinity.



Defective eye (A)

Corrected eye (B)

If P´ is far point for a myopic eye, then the image of an object placed at the point P´ will be formed on the retina as shown in the figure (A).

The myopic eye will get cured against this defect, if it is able to see the objects at infinity clearly. In order to correct the eye for this defect, a concave lens of suitable focal length is placed close to the eye, so that the parallel ray of light from point P´ of the myopic eye as shown in figure (B).

If x is the distance of the far point from the eye, then for the concave lens placed before the eye:

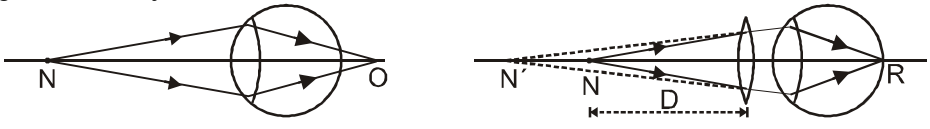
$u = \infty$ and $v = -x$

solving, $f = -x$

Thus, myopic eye is cured against the defect by using a concave lens of focal length equal to the distance of its far point from the eye.

2 Hypermetropia [Or Long-sightendness or far-sightendness]

In it near object are not clearly visible i.e., Near Point is at a distance greater than 25 cm and hence image of near object is formed behind the retina.



Defective Eye (A)

Corrected Eye (B)

In case of a hypermetropic eye, when the object lies at the point N (at the near point for a normal eye), its image is formed behind the retina as shown in figure (A).

The near point N´ for hypermetropic eye is farther than N, the near point for a normal eye.

Such defect will get cured, if the eye can see an object clearly, when place at the near point N for the normal eye. To correct this defect, a convex lens of suitable focal length is placed close to the eye so that the rays of light form an object placed at the point N after refraction through the lens appear to come from the near point N´ of the hypermetropic eye as shown in figure (B).

Let x be the distance of the near point N´ from the eye and D, the least distance of distinct vision i.e. the distance of near point N for the normal eye. Then, for the convex lens placed before the eye,

$u = -D$

and $v = -x$

If f is the focal length of the required convex lens, then from the lens formula, we have

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-x} - \frac{1}{-D} = \frac{1}{f}$$

$$f = \frac{xD}{x - D} \qquad ...(1)$$

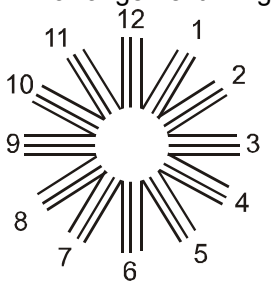
Thus, an eye suffering from hypermetropia can be cured against the defect by using a convex lens of focal length given by equation (1).

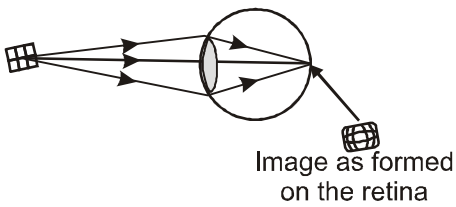
3 Presbyopia

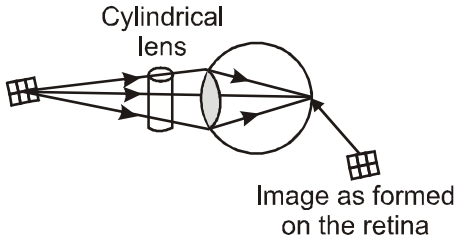
In this both near and far objects are not clearly visible i.e., far point is lesser than infinity and near point greater than 25 cm. It is an old age disease as at old age ciliary muscles lose their elasticity and so can not change the focal length of eye-lens effectively and hence eye loses its power of accommodation.

4 Astigmatism

In it due to imperfect spherical nature of eye-lens, the focal length of eye lens is two orthogonal directions becomes different and so eye cannot see object in two orthogonal directions clearly simultaneously. This defect is directional and is remedied by using cylindrical lens in particular direction. If in the spectacle of a person suffering from astigmatism, the lens is slightly rotated the arrangement will get spoiled.









Example 1. A person cannot see objects clearly beyond 50 cm. What should be the power of corrective lens used ?

Solution : $$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

for correcting for point $u = -\infty$, $v = -50$ cm

$$-\frac{1}{50} + \frac{1}{\infty} = \frac{1}{f}$$

$f = -50$ cm

$$P = \frac{1}{f} = \frac{1}{-0.5} = -2D$$

Example 2. A certain myopic person has a far point of 150 cm. (a) What power a corrective lens must have to allow him to see distant objects clearly ? (b) If he is able to read a book at 25 cm, while wearing the glasses, is his near point less than 25 cm ?

Solution : (a) Here, the distance of the far point, x = 150 cm
The defect can be corrected by using concave lens of focal length,
f = –x = –150 cm = –1.5 m
The power of the lens is given by

$$P = \frac{1}{f} = \frac{1}{-1.5} = 0.67 \text{ D}$$

(b) here, u = –25 cm ; f = –150 cm
From the lens equation, we have

$$v = \frac{uf}{u+f} = \frac{(-25)\times(-150)}{(-25)+(-150)} = 21.43 \text{ cm}$$

Therefore, the near point will be at a distance of 21.43 cm i.e. less than 25cm.

Optical instruments used primarily to assist the eye in viewing an object.

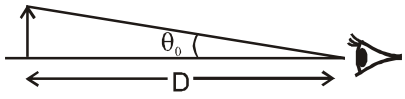
1. Microscope

It is an optical instrument used to increase the visual angle of neat objects which are too small to be seen by naked eye.

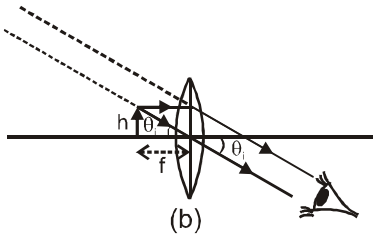
1.1 Simple Microscope

The normal human eye can focus a sharp image of an object on the retina if the object is located anywhere from infinity to a certain point called the **near point (D)**. If you move the object closer to the eye than the near point, the perceived retinal image becomes fuzzy. For an average viewer the near point, D = 25 cm from the eye. When the object is at the eye's near point, its image on the retina is as large as it can be and still be in focus.

The apparent size of an object is determined by the size of its image on the retina. If the eye is unaided, this size depends on the angle θ_o subtended by the object at the eye, called its **angular size** as shown in figure (a).



unaided eye
(a)



(b)

To look closely at a small object, such as an insect or a crystal, you bring it close to your eye, making the subtended angle and the retinal image as large as possible. But your eye cannot focus sharply on objects that are closer than the near point, so the angular size of an object is greatest (that is, it subtends the largest possible viewing angle) when it is placed at the near point.

A converging lens can be used to form a virtual image that is larger and farther from the eye than the object itself, as shown in figure (b). Then the object can be moved closer to the eye, and the angular size of the image may be substantially larger than the angular size of the object at 25 cm without the lens. A lens used in this way is called a **simple microscope,** otherwise known as a magnifying glass.

The usefulness of the magnifier is given by its angular magnification.

(i) I**f the image is formed at infinity (normal adjustment):** The virtual image is most comfortable to view when it is placed at infinity, so that the ciliary muscle of the eye is relaxed; this means that the object is placed at the focal point of the magnifier. In this case we find angular magnification.

Angular magnification or magnifying power (M) is defined as the ratio of the angle subtended by the image (situated at infinity) at the eye to the angle subtended by the object seen directly at the eye when situated at near point D.

In figure (a) the object is at the near point, where it subtends an angle θ_o at the eye.

$$\theta_o \approx \tan\theta_o \approx \frac{h}{D} \qquad ...(1)$$

In figure (b) a magnifier in front of the eye forms an image at infinity, and the angle subtended at the magnifier is θ_i

$$\theta_i \approx \tan\theta_i \approx \frac{h}{f} \qquad ...(2)$$

Angular magnification, $$M = \frac{\theta_i}{\theta_o} = \frac{D}{f} \qquad ...(3)$$

(ii) I**f the image is at formed at near point, D:** The linear magnification 'm', for the image formed at the near point D, by a simple microscope can be obtained by using the relation

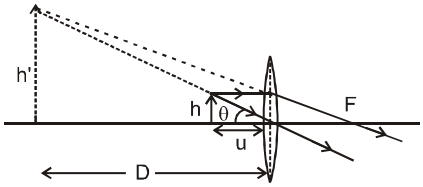
$$m = \frac{h'}{h} = \frac{v}{u} = \left(1 - \frac{v}{f}\right) \qquad ...(4)$$

h is the size of the object and h' is the size of the image

$$m = \left(1 + \frac{D}{f}\right) \qquad ...(5) \qquad (v = -D)$$

Angular magnification or magnifying power(M) is defined as the ratio of the angle subtended by the image at the eye to the angle subtended by the object seen directly at the eye when both lie at near point D.

In figure (c) a magnifier in front of the eye forms an image at D, and the angle subtended at the magnifier is θ_i



(c)

Example 3. A man with normal near point (25 cm) reads a book with small print using a magnifying thin convex lens of focal length 5 cm. (a) What is the closest and farthest distance at which he can read the book when viewing through the magnifying glass? (b) What is the maximum and minimum magnifying power possible using the above simple microscope?

Solution : (a) As for normal eye far and near point are and 25 cm respectively, so for magnifier $v_{max} = -\infty$ and $v_{min} = -25\,cm$. However, for a lens as

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \qquad \text{i.e.,} \qquad u = \frac{f}{\left(\frac{f}{v}\right) - 1}$$

So u will be minimum when $v_{min} = -25\,cm$

i.e., $(u)_{min} = \dfrac{5}{\left(\frac{-5}{25}\right) - 1} = -\dfrac{25}{6} = -4.17\,cm$

And u will be maximum when $v_{max} = \infty$

So, the closest and farthest distance of the book from the magnifier (or eye) for clear viewing are 4.17 cm and 5 cm respectively.

(b) An in case of simple magnifier $MP = \left(\frac{D}{u}\right)$. So MP will be minimum when u_{max} = 5 cm

i.e. $(MP)_{min} = \dfrac{-25}{-5} = 5 \qquad \left[= \dfrac{D}{f}\right]$

And MP will be maximum when $u_{min} = \left(\frac{25}{6}\right)cm$ i.e., $(MP)_{max} = \dfrac{-25}{-\left(\frac{25}{6}\right)} = 6\left[= 1 + \dfrac{D}{f}\right]$

1.2 Compound Microscope

When we need greater magnification than what we can get with a simple magnifier, the instrument that we usually use is a compound microscope.

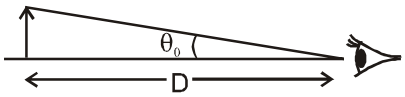
The essential parts of a compound microscope are two convex lenses of different focal length placed coaxially. These lenses are referred to as:

(a) Objective lens or objective: It is a lens of small aperture and small focal length placed facing the object.

(b) Eye piece: It is a lens of large aperture and small focal length placed facing the object.

The object O to be viewed is placed just beyond the first focal point of the **objective** lens that forms a real and enlarged image I′ as shown in figure. In a properly designed instrument this image lies just inside the first focal point of the **eyepiece**. The eyepiece acts as a simple magnifier, and forms a final virtual image of I. The position of may be anywhere between the near and far points of the eye.

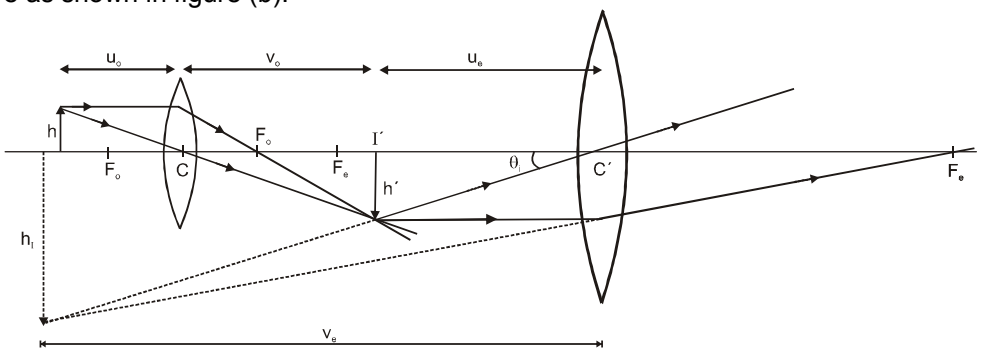
In figure (a) the object is at the near point, where it subtends an angle θ_o at the eye.



unaided eye
(a)

$$\theta_o \approx \tan\theta_o \approx \frac{h}{D} \qquad ...(1)$$

(i) When image is formed at near point, D: Let θ_i be the angle subtended by the final image at the eye as shown in figure (b).



(b)

Angular magnification or magnifying power (M) is defined as the ratio of the angle subtended by the final image at the eye to the angle subtended by the object seen directly at the eye when both lie at near point D.

The angular magnification produced is,

$$M = \frac{\theta_i}{\theta_o} \approx \frac{\tan\theta_i}{\tan\theta_o} \qquad ...(2)$$

$$\theta_i \approx \tan\theta_i = \frac{h_i}{v_e} = \frac{h_i}{D} \qquad (\because v_e = D \text{ in magnitude})$$

$$\theta_o \approx \tan\theta_o = \frac{h}{D}$$

$$M = \frac{h_i}{h} \qquad ...(3)$$

Linear magnification, $m = \frac{h_i}{h} = m_o \times m_e$ ...(4)

$M = m_o m_e$ (from eq. (3) and eq. (4))

where m_o = linear magnification produced by objective lens = $\frac{v_o}{u_o}$ ...(5)

m_e = linear magnification produced by eye piece = $\frac{v_e}{u_e}$

using lens formula for eye piece, $\frac{1}{v_e}-\frac{1}{u_e}=\frac{1}{f_e}$

$$m_e=\frac{v_e}{u_e}=1-\frac{v_e}{f_e}=1+\frac{D}{f_e} \qquad ...(6) \quad (\because v_e=-D)$$

From equations (3), (4) and (6) we have

$$M=\frac{v_o}{u_o}\left(1+\frac{D}{f_e}\right), \qquad ...(7)$$

In practice, the focal length of objective lens is very small and the object is just placed outside the focus of the objective lens,

$u_o = -f_o$

Since the focal length of the eye lens is also small, the distance of image I' from objective lens is nearly equal to the length of the microscope tube, L

$v_o = L$

substituting in equation (7),

$$M=-\frac{L}{f_o}\left(1+\frac{D}{f_e}\right)$$

This equation shows that a compound microscope will have high magnifying power, if the objective lens and the eye piece both have small focal length. The negative sign shows that final image will be inverted w.r.t. object.

(ii) **When image is formed at infinity**: The magnifying power of compound microscope is given by

$M = m_o \times m_e$

Magnification produced by objective lens, $m_o = \frac{v_o}{u_o}$

The eye lens produces the final image at infinity. Then,

$m_e = \frac{D}{f_e}$ (as discussed in case of simple microscope)

Therefore, $M=\frac{v_o}{u_o}\frac{D}{f_e}$,

$$M=-\frac{L}{f_o}\frac{D}{f_e}$$



Example 4. The focal length of the objective and eyepiece of a microscope are 2 cm and 5 cm respectively and the distance between them is 20 cm. Find the distance of object from the objective, when the final image seen by the eye is 25 cm from the eyepiece. Also find the magnifying power.

Solution : Given $f_0 = 2$ cm, $f_e = 5$ cm

$| v_o | + | u_e | = 20$ cm

$\therefore \quad v_e = -25$ cm

From lens formula $\frac{1}{f_e}=\frac{1}{v_o}-\frac{1}{u_e}$

$$\frac{1}{u}=\frac{1}{v_e}-\frac{1}{f_e}=-\frac{1}{25}-\frac{1}{5}$$

$\therefore \quad u_e = -\frac{25}{6}$ cm

Distance of real image from objective

$$v_o = 20 - |u_e| = 20 - \frac{25}{6} = \frac{120-25}{6} = \frac{95}{6} \text{ cm}$$

Now $$\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$$

given $$\frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{(95/6)} - \frac{1}{2} \quad \text{i.e.,} \quad \frac{1}{u_o} = \frac{6}{95} - \frac{1}{2} = \frac{12-95}{190} = -\frac{83}{190}$$

$$\therefore \quad u_o = -\frac{190}{83} = -2.3 \text{ cm}$$

Magnifying power $$M = -\frac{v_o}{u_o}\left(1+\frac{D}{f_e}\right) = -\frac{95/6}{(190/83)}\left(1+\frac{25}{3}\right) = -41.5$$

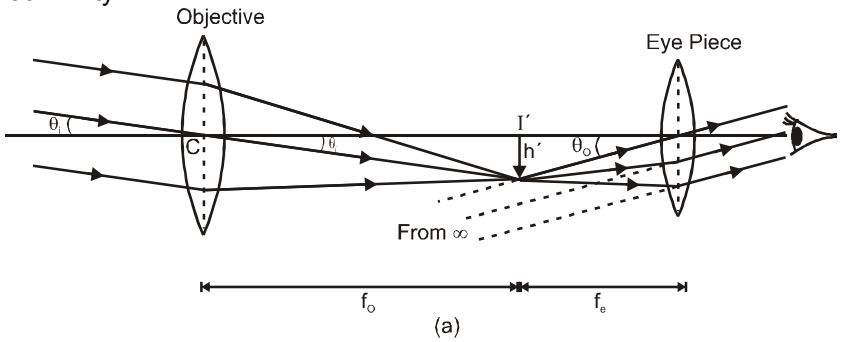
2. Telescope

2.1 Astronomical Telescope

It is an optical instrument used to increase the visual angle of distant large objects such as a star a planet or a cliff etc. Astronomical telescope consists of two converging lens. The one facing the object is called objective and has large focal length and aperture. Other lens is called eye piece. It has small aperture and is of small focal length. The distance between the two lenses is adjustable.
The objective forms a real and inverted image at its focal plane of the distant object. The distance of the eye piece is adjusted, till the final image is formed at the near point, D. In case, the position of the eye piece is so adjusted that final image is formed at infinity, the telescope is said to be in normal adjustment.

(i) When the image is formed at infinity (Normal adjustment)

When a parallel beam of light rays from a distant object falls on objective, its real and inverted image I′ is formed on the other side of the objective and at a distance f_o. If the position of the eye piece is adjusted, so that the image I′ lies at its focus, then the final highly magnified image will be formed at infinity.



(a)

Angular magnification or magnifying power (M) here is defined as the ratio of the angle subtended by the final image at the eye as seen through the telescope to the angle subtended by the object, seen directly at the eye when both the object and the image lie at infinity.

$$M = \frac{\theta_i}{\theta_o} = \frac{\tan\theta_i}{\tan\theta_o} \qquad \text{(for small angle } \tan\theta \approx \theta\text{)}$$

From figure (a), $\tan\theta_i = \frac{h'}{CI'}$, $\tan\theta_o = \frac{h'}{C'I'}$

$$M = \frac{CI'}{C'I'} = -\frac{f_o}{f_e} \qquad (CI' = f_o,\ C'I' = -f_e)$$

If f_o is large and f_e is small, the magnification will be high. In normal adjustment the length of tube

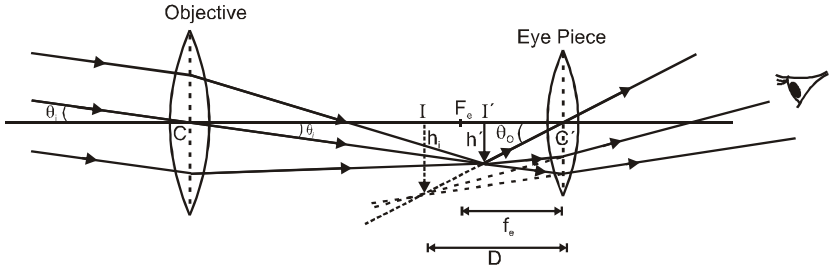
$$L = (f_0 + u_e)$$

(ii) If the final image is formed at D (near point)

When a parallel beam of light rays from a distant object falls on objective, its real and inverted image I′ is formed on the other side of the objective and at a distance f_o. If the position of the eye piece is adjusted, so that the final image I is formed at near point D.

Angular magnification or magnifying power (M) here is defined as the ratio of the angle subtended by the final image formed at near point at the eye to the angle subtended by the object lying at infinity seen directly at the eye.

$$M = \frac{\theta_i}{\theta_o} = \frac{\tan\theta_i}{\tan\theta_o} \qquad \text{(for small angle } \tan\theta \approx \theta\text{)}$$



(b)

From figure (b), $\tan\theta_o = \frac{h'}{CI'}$, $\tan\theta_i = \frac{h'}{C'I'}$

$$M = \frac{CI'}{C'I'} = -\frac{f_o}{u_e}$$

For eye lens, $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$,

$$\frac{1}{u_e} = \frac{1}{f_e} + \frac{1}{D} \qquad (\because\ u_e = -u_e,\ v_e = -D)$$

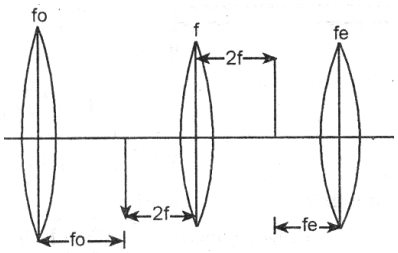
$$M = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$$

If f_o is large and f_e is small, the magnification will be high.

2.2 Terrestrial Telescope

Uses a third lens in between objective and eyepieces so as to form final image erect. This lens simply invert the image formed by objective without affecting the magnification.

Length of tube $L = f_0 + f_e + 4f$



Example 5. A telescope consists of two convex lens of focal length 16 cm and 2 cm. What is angular magnification of telescope for relaxed eye? What is the separation between the lenses? If object subtends an angle of 0.5º on the eye, what will be angle subtended by its image ?

Solution : Angular magnification

$$M = \frac{\alpha}{\beta} = \frac{F}{f} = \frac{16}{2} = 8 \text{ cm}$$

Separation between lenses = F + f = 16 + 2 = 18 cm

Here $\alpha = 0.5^\circ$

$\therefore$ Angular subtended by image

$\beta = M\alpha = 8 \times 0.5^\circ = 4^\circ$

Example 6. The magnifying power of the telescope if found to be 9 and the separation between the lenses is 20 cm for relaxed eye. What are the focal lengths of component lenses ?

Solution : Magnification $M = \frac{F}{f}$

Separation between lenses

$d = F + f$

Given $\frac{F}{f} = 9$ i.e., $F = 9f$ .....(1)

and $F + f = 20$ .....(2)

Putting value of F from (1) in (2), we get

$9f + f = 20 \Rightarrow 10\,f = 20 \Rightarrow f = \frac{20}{10} = 2\text{cm}$

$\therefore$ $F = 9f = 9 \times 2 = 18$ cm

$\therefore$ F = 18 cm, f = 2 cm

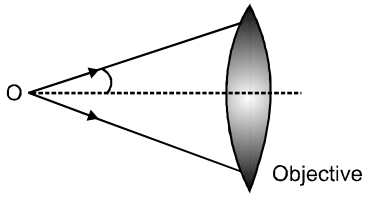
Comparison between Compound - Microscope & Astronomical - Telescope

S.No.	Compound - Microscope	Astronomical - Telescope
1.	It is used to increase visual angle of near tiny object.	It is used to increase visual angle of distant large objects.
2.	In it field and eye lens both are convergent, of short focal length and aperture. and	In it field lens is of large focal length and aperture while eye lens of short focal length aperture and both are convergent.
3.	Final image is inverted. virtual and enlarged and at a distance D to ∞ from the eye.	Final image in inverted, virtual and enlarged at a distance D to ∞ from the eye
4.	MP does not change appreciably if field and eye lens are interchanged [MP ~ $(LD/f_0\, f_e)$] ~$[f_0/f_e]$	MP becomes $(1/m^2)$ times of its initial value if field and eye-lenses are interchanged as MP
5.	MP is increased by decreasing the focal of length of both the lenses viz. find and length eye lens.	MP is increased by increasing the focal length field of field lens (and decreasing the focal of eye lens.)
6.	RP is increased by decreasing the wavelength of light used.	RP is increased by increasing the aperture of objective.

RESOLVING POWER (R.P.)

(1) Microscope : In reference to a microscope, the minimum distance between two lines at which they are just distinct is called Resolving limit (RL) and it's reciprocal is called Resolving power (RPO)

$\text{R.L.} = \frac{\lambda}{2\mu\sin\theta}$ and $\text{R.P.} = \frac{2\mu\sin\theta}{\lambda} \Rightarrow \text{R.P.} \propto \frac{1}{\lambda}$



λ = Wavelength of light used to illuminate the object

μ = Refractive index of the medium between object and objective.

θ = Half angle of the cone of light from the point object, $\mu \sin\theta$ = Numerical aperture.

(2) Telescope : Smallest angular separations ($d\theta$) between two distant object, whose images are separated in the telescope is called resolving limit. So resolving limit $d\theta = \frac{1.22\lambda}{a}$ and resolving power

$(RP) = \frac{1}{d\theta} = \frac{a}{1.22\lambda} \Rightarrow \text{R.P.} \propto \frac{1}{\lambda}$ where a = aperture of objective.

Example 7 Find the half angular width of the central bright maximum in the Fraunhofer diffraction pattern of a slit of width 12×10^{-5} cm when the slit is illuminated by monochromatic light of wavelength 6000Å.

Solution. Here $\sin\theta = \frac{\lambda}{a}$

Where θ is half angular width of the central maximum.

$A = 12 \times 10^{-5}$ cm, $\lambda = 6000Å = 6 \times 10^{-5}$cm.

$\therefore \quad \sin\theta = \frac{\lambda}{a} = \frac{6 \times 10^{-5}}{12 \times 10^{-5}} = 0.50$

or $\theta = 30°$

Example 8 In Fraunhofer diffraction due to a narrow slit a screen is placed 2 m away from the lens to obtain the pattern. If the slit width is 0.2 mm and the first minima lie 5 mm on either side of the central maximum, find the wavelength of light.

Solution. In the case of Fraunhofer diffraction at a narrow rectangular aperature,

$a \sin\theta = n\lambda$

$n = 1$

$\therefore \quad a \sin\theta = \lambda$

$\sin\theta = \frac{x}{D}$

$\therefore \quad \frac{ax}{D} = \lambda$

$\lambda = \frac{ax}{D}$

Here $a = 0.2$ mm $= 00.2$cm

$x = 5$ mm $= 0.5$ cm

$D = 2$m $= 200$ cm

$\therefore \quad \lambda = \frac{0.02 \times 0.5}{200}$

$\lambda = 5 \times 10^{-5}$ cm

$L = 5000Å$

Example 9. Light of wavelength 6000Å is incident on a slit of width 0.30 nm. The screen is placed 2m from the slit. Find : (a) the position of the first dark fringe and (b) the width of the central bright fringe.

Solution. The first dark fringe is on either side of the central bright fringe.

Here, $n = \pm 1$, $D = 2$m

$\lambda = 6000Å = 6 \times 10^{-7}$m

$\sin\theta = \frac{x}{D}$

$a = 0.30$ mm $= 3 \times 10^{-4}$ m

$a \sin\theta = n\lambda$

(a) $x = \frac{n\lambda D}{a}$

$x = \pm \left[\frac{1 \times 6 \times 10^{-7} \times 2}{3 \times 10^{-4}}\right]$

$x = \pm 4 \times 10^{-3}$m

The positive and negative signs correspond to the dark fringes on either side of the central bright fringe.

(b) The width of the central bright fringe,

$y = 2x$

$= 2 \times 4 \times 10^{-3}$

$= 8 \times 10^{-3}$ m $= 8$ mm

Example 10. A signal slit of width 0.14 mm is illuminated normally by monochromatic light and diffraction bands are observed on a screen 2m away. If the centre of the second dark band is 1.6 cm from the middle of the central bright band, deduce the wavelength of light used.

Solution. In the case of Fraunhofer diffraction at a narrow rectangular slit,

$a \sin\theta = n\lambda$

Here θ gives the directions of the minimum

$n = 2$, $\lambda = ?$

$a = 0.14 \text{ mm} = 0.14 \times 10^{-3} \text{ m}$

$D = 2 \text{ m}$

$x = 1.6 \text{ cm} = 1.6 \times 10^{-2} \text{ m}$

$\sin\theta = \frac{x}{D} = \frac{n\lambda}{a} \therefore \quad \lambda = \frac{xa}{nD}$

$= \frac{1.6\times10^{-2}\times0.14\times10^{-3}}{2.2} = 5.6 \times 10^{-7} \text{ m} = 5600\text{Å}$

Example 11. A screen is placed 2m away from a narrow slit which is illuminated with light of wavelength 6000Å. If the first minimum lies 50 mm on either side of the central maximum, calculate the slit width.

Solution. In the case of Fraunhofer diffraction at a narrow slit,

$a \sin\theta = n\lambda$

$\sin\theta = \frac{x}{D} \quad \therefore \quad \frac{ax}{D} = n\lambda$

Here width of the slit = $a = ?$

$x = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$

$D = 2\text{m}$

$\lambda = 6000\text{Å} = 6 \times 10^{-7}\text{m}$

$n = 1$

$a = \left(\frac{n\lambda D}{x}\right)$

$a = \left(\frac{1\times6\times10^{-7}\times2}{5\times10^{-3}}\right)$

$a = 2.4 \times 10^{-4} \text{ m}$

$a = 0.24 \text{ mm}$

Example 12. Find the angular width of the central bright maximum in the Fraunhofer diffraction pattern of a slit of width 12×10^{-5} cm when the slit is illuminated by monochromatic light of wavelength 6000Å.

Solution. Here $\sin\theta = \frac{\lambda}{a}$

where θ is the half angular width of the central maximum

$a = 12 \times 10^{-5} \text{ cm} = 12 \times 10^{-7} \text{ m}$

$\lambda = 6000\text{Å} = 6 \times 10^{-7} \text{ m}$

$\sin\theta = \frac{6\times10^{-7}}{12\times10^{-7}} = 0.5$

$\theta = 30°$

Angular width of the central maximum.

$2\theta = 60°$

Example 13. Diffraction pattern of a signal slit of width 0.5 cm is formed by a lens of focal length 40 cm. Calculate the distance between the first dark and the next bright fringe from the axis. Wave length = 4890Å.

Solution. For minimum intensity

$a \sin \theta_n = n\lambda$

$\sin \theta_n = \frac{x_1}{f'} \Rightarrow n = 1$

$\frac{x_1}{f} = \frac{\lambda}{a}$

Here $\lambda = 4890Å = 4890 \times 10^{-10}$ m

$a = 0.5$ cm $= 5 \times 10^{-3}$ m

$f = 40$ cm $= 0.4$ m

$x_1 = \frac{f\lambda}{a}$

$x_1 =$

$x_1 = 3.912 \times 10^{-5}$ m

For secondary maximum

$a \sin \theta_n = \frac{0.4 \times 4890 \times 10^{-10}}{5 \times 10^{-3}}$

For the first secondary maximum

$n = 1$

$\sin \theta_n = \frac{(2n+1)\lambda}{2}$

$\frac{x_2}{f} = \frac{3\lambda}{2a}$

$x_2 = \frac{3\lambda f}{2a}$

$x_2 = \frac{3 \times 4890 \times 10^{-10} \times 0.4}{2 \times 5 \times 10^{-5}}$

$x_2 = 5.868 \times 10^{-5}$ m

Difference,

$x_2 - x_1 = 5.868 \times 10^{-5} - 3.912 \times 10^{-5}$

$= 1.956 \times 10^{-5}$ m

$= 1.596 \times 10^{-2}$ mm

Example 14. Find the separation of two points on the moon that can be resolved by a 500 cm telescope. The distance of the moon is 3.8×10^5 km. The eye is most sensitive to light of wavelength 5500 Å.

Solution. The limit of resolution of a telescope is given by

$d\theta = \frac{1.22\lambda}{a}$

Here $\lambda = 5500 \times 10^{-8}$cm, $a = 500$ cm

$\therefore \quad d\theta = \frac{1.22 \times 5500 \times 10^{-8}}{500}$

$\therefore \quad d\theta = 13.42 \times 10^{-8}$ radian

Let the distance between the two point be x

$\therefore \quad d\theta = \frac{x}{R}$

Here $R = 3.8 \times 10^{10}$ cm

$x = R.d\theta$

$= 3.8 \times 10^{10} \times 13.42 \times 10^{-8}$

$= 50.996 \times 10^2$ cm $= 50.996$ meters

Example 15. Calculate the aperture of the objective of a telescope which may by used to resolve stars separated by 4.88×10^{-6} radian for light of wavelength 6000Å

Solution. Here $\lambda = 6000Å = 6 \times 10^{-5}$ cm, $\theta = 4.88 \times 10^{-6}$ radian

D = ?

$$\theta = \frac{1.22\lambda}{\theta}$$

or $$D = \frac{1.22\lambda}{\theta} = \frac{1.22 \times 6 \times 10^{-5}}{4.88 \times 10^{-8}} = 15\text{cm}$$

Example 16. Two pin holes 1.5 mm apart are placed in front of a source of light of wavelength 5.5×10^{-5} cm and seen through a telescope with its objectives stopped down to a diameter of 0.4 cm. Find the maximum distance from the telescope at which the pin holes can be resolved.

Solution. Here, $\lambda = 5.5 \times 10^{-5}$ cm

a = 0.4 cm

$$d\theta = \frac{1.22\lambda}{a}$$

Also $$d\theta = \frac{x}{d}$$

x = 1.5 mm = 0.15 cm

$$\therefore \quad \frac{x}{d} = \frac{1.22\lambda}{a}$$

$$d = \frac{xa}{1.22\lambda}$$

$$d = \frac{0.15 \times 0.4}{1.22 \times 5.5 \times 10^{-5}}\text{cm}$$

d = 894.2 cm = 8.942 m